

the $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by

$$\|\mathcal{H}_\infty\| \leq \sqrt{\lambda_{\max}(\mathbf{P})} \quad (10)$$

where $\mathbf{P}$ is the solution of the Lyapunov equation

$$\mathbf{A}^T \mathbf{P} + \mathbf{P} \mathbf{A} + \mathbf{C}^T \mathbf{C} = 0 \quad (11)$$

and $\lambda_{\max}(\mathbf{P})$ is the maximum eigenvalue of $\mathbf{P}$.

It is worth noting that the $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.

The $\mathcal{H}_\infty$ norm of the closed-loop system is bounded by the $\mathcal{H}_\infty$ norm of the plant.